

James Edralin

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TECHNICAL SKILLS

Languages, Graphics API, and Libraries : C++, Python, GLSL, HLSL, OpenGL, GLFW, Assimp, ImGui
Unreal Engine 5 : Blueprints, Material Editor, PCG, Editor Utilities, Scriptable Tools, XR Development, Insights
Engine and Rendering : ECS Architecture, Deferred Rendering, Shader Development, Terrain Systems
Version Control and Tooling: Perforce, Git, CMake, RenderDoc

WORK EXPERIENCE

Software Developer, UBC Emerging Media Lab May 2024 – Present

- Built a VR sound localization evaluation application using Blueprints, PCG, Niagara Modules, and engine subsystems, presented in formal academic research to evaluate localization accuracy across sensory cues
- Developed procedural gameplay systems and integrated AI semantic-analysis outputs into environment generation for a literature-exploration prototype, securing \$100K+ CAD in federal research funding
- Extended Unreal Editor workflows with scriptable tools, inspector utilities, and modular UI, improving developer efficiency (-30% level-composition time)
- Authored custom lighting models, post-processing passes, and material workflows (instances, functions, collections), optimizing real-time rendering for VR/XR applications

VR Developer, EarSim Digital, UBC School of Audiology and Speech Sciences Jun 2025 – Sep 2025

- Developed a standalone C++/ImGui data calibration toolkit for live-tune spatial audio parameters and set up randomized block design experiments to a VR application
- Implemented engine subsystems with JSON-driven import/export pipelines, handling serialization, GUI-authored configs, and Unreal parsing to automate experiment setup and data processing
- Built automated validation and functional test workflows; instrumented telemetry logging and auto-generated reports to ensure reproducibility and reduce setup errors

PROJECTS

PBR Graphics Engine | C++, OpenGL, GLSL, ImGui, RenderDoc, CMake Feb 2025 – Jun 2025

- Developed a modular deferred rendering engine, supporting physically-based rendering (PBR), bloom, tone mapping, HDR, and customizable post-processing render passes. Each visualized directly in a built-in debugger inspector
- Built a custom ECS-based architecture for managing of scenes, assets, materials, and render resources
- Implemented a tiled-based lighting system using compute shaders to handle multiple dynamic lights efficiently
- Added dynamic image-based lighting (IBL) with nearest-probe selection and blending for accurate spatial lighting
- Built terrain rendering with support for geomipmapping and tessellation-based level of detail (LOD)

3D Pixel Art Pipeline | Unreal Engine 5, Blueprints, Material Editor, PCG, Blender Nov 2024 – Jan 2025

- Authored a stylization workflow using compounded shader passes (pixelation, quantization, and outlining methods)
- Developed real-time reflections for water accomplished by dynamic render targets and distance fields
- Achieved real-time crepuscular ray effects using a custom 2D ray marching method and fractal brownian motion

PUBLICATIONS

Procedural Multi-sensory VR System Evaluation Research Poster Aug 2025

Third Place in Student Research Competition (ACM SRC 2025) - Undergraduate Category ACM, SIGGRAPH

EDUCATION

University of British Columbia Sep 2022 – May 2027

Bachelor of Computer Science (BCS Program) Vancouver, BC

Scholarship Award Recipient, First Place: Kirk Karasin Memorial Award (2025)

De La Salle – College of Saint Benilde, Design and Arts Aug 2015 – May 2020

Bachelor of Science in Architecture Metro Manila